

WAVE SHOES SALES ANALYTICS DASHBOARD

Wave Shoes — Enterprise Sales Intelligence Platform

Project Documentation Report

Power BI | Looker Studio | SQL | Python | Excel | Generative AI

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<https://github.com/deepak10281/Gen-AI-Powered-Sales-Analytics-Dashboard>

1. Executive Summary

"Wave Shoes delivered outstanding financial performance with ₹113M in total revenue, ₹25M in profit, and a solid 22.4% profit margin — powered by data-driven intelligence across Power BI, Looker Studio, SQL, Python, Excel, and Generative AI."

The Wave Shoes Sales Analytics Dashboard project represents a comprehensive, end-to-end business intelligence solution built for Wave Shoes — an Indian footwear brand. This project integrates multiple modern analytics tools and AI technologies to deliver actionable insights across sales, profitability, customer behavior, and regional performance.

The project was architected with an enterprise-grade mindset, combining structured SQL analytics with Python-based exploratory data analysis, multi-platform BI dashboards, and Generative AI to surface nuanced business intelligence. The resulting platform empowers decision-makers with real-time insights, strategic forecasting, and deep product and customer analytics.

Total Revenue	Total Profit	Profit Margin	Total Quantity	Transactions	Sales Growth
₹113M	₹25M	22.4%	24K	7.8K	4.76%

Key highlights of the project include:

- Comprehensive sales intelligence across 30 Indian states and union territories
- Product performance tracking across 6 shoe types in 3 categories
- Month-over-month trend analysis revealing seasonal demand patterns
- AI-augmented narrative reporting using Generative AI models
- Multi-platform dashboard deployment: Power BI, Google Looker Studio, and Excel
- Advanced SQL analytics using CTEs, Window Functions, and ranking queries

2. Business Problem Statement

Wave Shoes, a fast-growing Indian footwear brand, faced critical challenges in extracting meaningful insights from its transactional sales data. Despite consistent revenue generation, the business lacked visibility into:

- Which product categories and specific shoe types were driving the most revenue and profit
- How sales trended month-over-month and whether growth was sustainable
- Which geographic markets (states and cities) represented the highest opportunity
- Where margin erosion was occurring across products and regions
- The behavioral patterns of high-value customers and their purchase frequency
- The performance differential between online and offline sales channels

Without a unified analytics layer, leadership was making strategic decisions based on fragmented reports, spreadsheets, and intuition — leading to missed opportunities in inventory planning, regional expansion, and targeted marketing.

This project was designed to solve that problem by building a centralized, AI-augmented analytics platform that transforms raw transactional data into boardroom-ready business intelligence.

3. Project Objectives

This project was driven by the following strategic and technical objectives:

Strategic Objectives

1. Build an enterprise-grade, multi-platform analytics dashboard for Wave Shoes.
2. Enable leadership to monitor KPIs in real-time through interactive BI tools.
3. Identify underperforming markets and products for targeted intervention.
4. Forecast future sales trends using historical data patterns.
5. Deliver AI-powered narrative reporting for non-technical stakeholders.

Technical Objectives

6. Design a clean, normalized fact table (Fact_Sales) for analytics.
7. Write 15+ SQL queries using advanced concepts: CTEs, Window Functions, Ranking.
8. Perform Exploratory Data Analysis (EDA) and statistical profiling in Python.
9. Build interactive, filter-driven dashboards in Power BI and Looker Studio.
10. Create automated Excel reports with pivot tables and dynamic charts.
11. Integrate Generative AI to auto-generate business narrative summaries.

4. Dataset Overview

The project is built on a single, denormalized Fact_Sales table that captures all transactional sales data for Wave Shoes across India. The dataset was purpose-built to support multi-dimensional analytics across time, geography, product, customer, and channel dimensions.

Table: Fact_Sales

Column Name	Data Type	Description
order_number	VARCHAR	Unique identifier for each transaction
order_date	DATE	Date on which the order was placed
customer_name	VARCHAR	Full name of the purchasing customer
city	VARCHAR	City where the order was placed or delivered
state	VARCHAR	Indian state of the customer/delivery location
order_type	VARCHAR	Sales channel: Online or Offline
product	VARCHAR	Name of the shoe product (e.g., Loafers, Brogues)
category	VARCHAR	Product category: Casual, Formal, Semi-Casual
quantity	INT	Number of units sold per transaction
cost_per_unit	DECIMAL	Unit cost price of the product (in INR)
selling_price_per_unit	DECIMAL	Unit selling price of the product (in INR)
sales_amount	DECIMAL	Total revenue from the transaction (quantity x price)
cost	DECIMAL	Total cost for the transaction (quantity x cost per unit)
profit	DECIMAL	Gross profit (sales_amount - cost)
profit_margin_%	DECIMAL	Profit as a percentage of sales amount

Dataset Statistics

Attribute	Value
Total Records	7,826 transactions
Date Range	Full calendar year (January – December)
Geographic Coverage	30 Indian states and union territories
Product Count	6 unique shoe types
Product Categories	3 (Casual, Formal, Semi-Casual)
Sales Channels	2 (Online, Offline)

Total Revenue	₹113 Million
Total Profit	₹25 Million
Total Units Sold	24K units

5. Data Cleaning Process

Before any analysis, a rigorous data quality assessment was conducted. The cleaning process followed a structured pipeline to ensure analytical accuracy and data integrity:

Step 1: Initial Data Profiling

- Performed row count verification against source records
- Checked column completeness — all 15 columns validated for expected data types
- Identified and documented null values across all fields
- Detected and removed duplicate order_number entries

Step 2: Data Type Standardization

- Converted order_date from VARCHAR to proper DATE format (YYYY-MM-DD)
- Standardized decimal precision for financial columns (cost, sales_amount, profit)
- Normalized text fields: trimmed whitespace in customer_name, city, state, product
- Ensured category values were restricted to: Casual, Formal, Semi-Casual

Step 3: Business Rule Validation

- Validated: $\text{profit} = \text{sales_amount} - \text{cost}$ (flagged and corrected mismatches)
- Validated: $\text{profit_margin_}\% = (\text{profit} / \text{sales_amount}) * 100$
- Ensured $\text{selling_price_per_unit} > \text{cost_per_unit}$ for all records
- Verified quantity values were positive integers (no zero or negative orders)

Step 4: Outlier Detection

- Used Python (IQR method) to identify statistical outliers in sales_amount and profit
- Investigated outlier records — most were large bulk orders (legitimate, retained)
- Flagged extreme values in profit_margin_% exceeding 3 standard deviations

6. Data Transformation Steps

Post-cleaning, the data was enriched and transformed to support multi-dimensional analysis:

- Extracted temporal dimensions: Year, Month Number, Month Name, Week Number, Weekday Name from order_date
- Created a Month-Year column (YYYY-MM format) for time series aggregation
- Engineered a Revenue Band column: categorizing transactions into Low / Mid / High value
- Derived a Profit Category: Profitable / Break-Even / Loss (based on profit margin thresholds)
- Added a Quarter column (Q1, Q2, Q3, Q4) for quarterly performance comparison
- Calculated Running Total Revenue (cumulative sum) for trend visualization
- Computed Month-over-Month (MoM) growth percentage for revenue and profit
- Added a Days Since Order column for recency analysis in customer segmentation
- Normalized state names for consistent mapping with geographic visualizations

7. SQL Analytics Performed

A total of 15 SQL analytical queries were developed to answer critical business questions across 5 analytical domains:

Business Performance Queries

12. Overall Business Performance — Total revenue, profit, margin, quantity, customer count, and product count from a single aggregate query.
13. Monthly Revenue & Profit Trend — Month-wise breakdown of revenue, profit, and average margin using DATE_FORMAT grouping.
14. Daily Sales Performance — Day-level granularity for identifying micro-trends and high-revenue days.
15. Weekday Sales Analysis — DAYNAME()-based aggregation revealing which days of the week generate the most sales.

Product & Category Analytics

16. Top Selling Products — Top 10 products by revenue with category classification.
17. Top Profitable Products — Products ranked by gross profit and average margin.
18. Low Profit Products — Identifies products with the lowest average margin requiring pricing review.
19. Category Performance — Aggregated cost, sales, and profit by category (Casual, Formal, Semi-Casual).
20. Sales Contribution by Category — Percentage share of each category in total revenue.

21. Top 5 Products in Each Category — Category-partitioned product ranking using RANK() window function.

Customer & Regional Analytics

22. Best Customers — Top 10 customers by spend with order count and city/state details.
23. City-Wise Sales Analysis — Revenue, profit, and customer count by city and state.

Channel & Growth Analytics

24. Online vs. Offline Sales — Comparative analysis of both channels: orders, revenue, cost, and profit.
25. Month-Over-Month Growth — Revenue growth percentage between consecutive months.
26. Running Sales Trend — Cumulative revenue progression across the year.

8. Advanced SQL Concepts Used

Common Table Expressions (CTEs)

CTEs were used to modularize complex queries, particularly in Month-over-Month growth and Running Sales Trend analyses. CTEs improved query readability and enabled referencing aggregated results in the outer SELECT.

```
Example: WITH monthly_sales AS (SELECT DATE_FORMAT(order_date,'%Y-%m') AS month_,  
SUM(sales_amount) AS revenue FROM Fact_Sales GROUP BY month_)  
SELECT month_, revenue, SUM(revenue) OVER(ORDER BY month_) AS running_revenue  
FROM monthly_sales;
```

Window Functions

- LAG() — Used in MoM growth to retrieve previous month revenue for variance calculation
- SUM() OVER() — Running cumulative total of revenue across months
- RANK() OVER(PARTITION BY) — Top-N products within each category using category-level partitioning

Aggregate Functions & Grouping

- SUM(), AVG(), COUNT(), ROUND() — Core aggregation for KPI calculation
- GROUP BY with multi-column grouping (product + category, city + state)
- ORDER BY with DESC/ASC for ranking
- LIMIT for top-N result sets

Date Functions

- DATE_FORMAT(order_date, '%Y-%m') — Monthly bucketing for trend analysis
- DAYNAME(order_date) — Weekday extraction for behavioral analysis

Subqueries

- Correlated subquery for calculating category percentage contribution relative to total revenue

9. Python Analysis Workflow

Python was used for Exploratory Data Analysis (EDA), statistical profiling, and advanced visualization. The workflow was structured as a Jupyter Notebook (Wave_Shoes_Sales_Analytics.ipynb) organized into 6 analytical phases:

Phase 1: Data Loading & Inspection

- Loaded dataset using pandas read_csv / read_excel
- Inspected shape, dtypes, info(), describe() for statistical overview
- Identified null counts, unique value distributions, and memory usage

Phase 2: Data Cleaning in Python

- Applied strip() to remove whitespace in categorical text columns
- Converted order_date to datetime64 using pd.to_datetime()
- Dropped duplicate rows based on order_number
- Validated calculated columns: profit, profit_margin_%

Phase 3: Feature Engineering

- Extracted: month, month_name, quarter, weekday, year from order_date
- Created revenue_band using pd.cut() into Low / Mid / High segments
- Derived profit_category based on margin thresholds

Phase 4: Exploratory Data Analysis

- Univariate analysis: Distribution plots for quantity, sales_amount, profit
- Bivariate analysis: Revenue vs. profit_margin scatter, category vs. revenue bar charts
- Correlation heatmap of numeric columns
- Monthly trend line chart with moving averages

Phase 5: Statistical Analysis

- Outlier detection using IQR method and Z-score
- Descriptive statistics: mean, median, std deviation for key metrics
- Sales channel comparative statistics (online vs. offline)

Phase 6: AI-Powered Insights

- Integrated Generative AI API to generate automated business narrative summaries
- Fed aggregated data as prompts to generate executive-level commentary

- Produced AI-written insights per category, region, and product performance

10. Power BI Dashboard Development

The Power BI dashboard was designed as the primary executive reporting layer, providing an interactive, filterable view of all key business metrics across multiple visual dimensions.

Dashboard Architecture

- Single-page executive dashboard with 6 KPI metric cards
- 5 interactive slicers: Category, State, Customer Name, Order Type, City
- 7 core visualizations covering performance, trends, and geography

Visuals Implemented

Visual Type	Purpose	Dimensions
KPI Cards (6)	Display headline metrics at a glance	Revenue, Cost, Profit, Qty, Margin%, Growth%
Line Chart	Monthly revenue and profit trend	X: Month, Y: Sales Amount
Donut Chart	Sales distribution by product type	Segments: 6 product types with % labels
Horizontal Bar Chart	Quantity sold by category	Bars: Casual, Formal, Semi-Casual
Map Visual (Bing Maps)	Geographic sales distribution by state	Bubble size: Revenue, Color: Category
Table / Matrix	State-wise profit comparison with totals	Columns: State, Orders, Profit, Margin%
Top 10 States Filter	Filtered view of highest-revenue states	Sorted by total sales descending

Interactivity Features

- Cross-filtering: Clicking any visual filters all other visuals automatically
- Drill-down capability on the monthly trend line for granular date analysis
- Slicer synchronization across all 5 filter dimensions
- Tooltip enhancements showing additional metrics on hover

11. Looker Studio Dashboard Development

Google Looker Studio was used to build the secondary web-based dashboard, optimized for sharing and real-time collaboration. The dashboard mirrors key Power BI visuals while leveraging Looker Studio's native features for accessibility and embeddability.

Dashboard Components

- Header section with branding, logo, and 5 KPI metric scorecards
- Dropdown filters: Category, State, Order Type, City, Customer Name search box
- Google Maps integration for choropleth-style state-level sales mapping
- Donut chart for product-type sales distribution (Single Monk leads at 35.3%)
- Horizontal bar chart for category quantity comparison
- Line chart for monthly sales trend (March peak: ₹20.9M)
- Paginated state-profit table (top 30 states, sorted by profit descending)

Key Differences from Power BI Version

Looker Studio Feature	Power BI Equivalent
Native Google Maps integration	Bing Maps
Web-embeddable, shareable URL	Desktop-first app
Real-time Google Sheets connector	Direct data import
Customer name text search filter	Dropdown slicer
Paginated state table (6-10/30 view)	Full table scroll

12. Excel Dashboard Reporting

Microsoft Excel was used to create a structured, portable reporting layer — ideal for non-technical stakeholders who prefer spreadsheet-format reports:

- Pivot Tables for monthly revenue/profit summary, category breakdown, and state-level analysis
- Dynamic charts: clustered bar charts, line charts, and pie charts linked to pivot tables
- Conditional formatting for profit margin heatmaps across product rows
- SUMIF, COUNTIF, AVERAGEIF formulas for dynamic KPI calculation
- Named ranges and data validation for filter dropdown interactivity
- Slicers connected to pivot tables for one-click filtering by category, month, and state
- Dashboard layout with frozen header row, consistent color coding, and print-ready formatting

13. KPI Definitions

KPI	Formula / Definition	Value	Context
Total Sales	SUM(sales_amount)	₹113M	Total gross revenue across all transactions
Total Cost	SUM(cost)	₹88M	Estimated total cost based on consolidated project KPI reporting
Total Profit	SUM(profit)	₹25M	Gross profit after deducting costs
Profit Margin %	AVG(profit_margin_%)	22.4%	Healthy margin for retail footwear
Total Quantity	SUM(quantity)	24K units	Total units sold across all products
Total Transactions	COUNT(order_number)	7.8K	Total unique sales orders
Sales Growth %	(Current - Previous) / Previous * 100	4.76%	Year-over-year or MoM growth
Max Sales (Single Order)	MAX(sales_amount)	₹42,000	Highest value single transaction
Avg Order Value	Total Revenue / Transactions	~₹14.5K	Derived KPI for basket size analysis

14. DAX Measures Explanation

The following DAX (Data Analysis Expressions) measures were implemented in Power BI to power the KPI cards, trend analysis, and comparative visuals:

Core Financial Measures

```
Total Sales = SUM(Fact_Sales[sales_amount])
Total Cost = SUM(Fact_Sales[cost])
Total Profit = SUM(Fact_Sales[profit])
Avg Profit Margin % = AVERAGE(Fact_Sales[profit_margin_%])
Total Quantity = SUM(Fact_Sales[quantity])
Total Transactions = COUNTROWS(Fact_Sales)
```

Growth & Trend Measures

```
MoM Revenue Growth % =
VAR CurrentMonth = [Total Sales]
VAR PreviousMonth = CALCULATE([Total Sales],
    DATEADD(Fact_Sales[order_date], -1, MONTH))
RETURN DIVIDE(CurrentMonth - PreviousMonth, PreviousMonth, 0) * 100

Running Total Revenue =
CALCULATE([Total Sales],
    FILTER(ALL(Fact_Sales[order_date]),
        Fact_Sales[order_date] <= MAX(Fact_Sales[order_date])))
```

Ranking & Comparative Measures

```
Product Revenue Rank =
RANKX(ALL(Fact_Sales[product]), [Total Sales],, DESC, DENSE)

Category Sales % =
DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Fact_Sales[category]))) * 100
```

15. Business Insights

Revenue & Profitability

- Wave Shoes generated ₹113M in total sales with ₹25M in total profit
- A 22.4% profit margin is above the industry average for Indian retail footwear (~18-20%), indicating efficient cost management
- Sales growth of 4.76% demonstrates consistent, sustainable business expansion

Product Insights

- Single Monk shoes dominate revenue at 35.3% of total product sales — a clear hero product
- Oxford Shoes contribute 17.2%, making it the second-most popular product
- Derby Shoes (14.5%), Brogues (13.6%), Loafers (9.7%), and Sneakers (9.6%) complete the portfolio
- Casual category leads in quantity sold (11.3K units), followed by Formal (10.3K) and Semi-Casual (2.5K)

Seasonal Insights

- March is the highest revenue month at ₹20.9M — likely driven by pre-summer fashion cycles
- April records ₹17.8M, sustaining the spring demand peak
- November rebounds to ₹15.2M — driven by festive season (Diwali aftermath and wedding season)
- June and July are the weakest months (₹1.9M and ₹1.8M), indicating a clear monsoon slump

Customer Insights

- High concentration of repeat buyers in top revenue states
- Andhra Pradesh leads with 661 orders — the most active customer market
- Top customers exhibit multi-category purchasing behavior across both Casual and Formal lines

16. Trend Analysis

Monthly Sales Trend

Month	Sales Amount	Trend	Insight
March	₹20.9M	Peak	Highest month — pre-summer demand surge
April	₹17.8M	High	Strong follow-through from March peak
September	₹15.2M	High	Post-monsoon festive season recovery
November	₹11.8M	Moderate	Festive & wedding season boost
February	₹11.6M	Moderate	Valentine's season gifting uplift
December	₹9.9M	Moderate	Year-end clearance and holiday buying
May	₹7.8M	Declining	Heat wave impact on footfall
August	₹6.5M	Low	Monsoon slowdown continuing
October	₹5.1M	Low	Post-Navratri purchasing pause
January	₹3.2M	Low	Post-holiday spending fatigue
June	₹1.9M	Trough	Monsoon season — weakest period
July	₹1.8M	Trough	Lowest month — full monsoon impact

MoM Growth Pattern

- Strongest MoM growth transitions: January to February, August to September (seasonal recoveries)
- Sharpest MoM declines: April to May, November to December
- The business follows a bimodal seasonal pattern with peaks in Q1 and Q3

17. Product Performance Analysis

By Revenue (Top Products)

Product	Category	Revenue Share	Performance Tier
Single Monk	Formal	35.3%	S-Tier (Hero Product)
Oxford Shoes	Formal	17.2%	A-Tier (Core Performer)
Derby Shoes	Semi-Casual	14.5%	A-Tier (Solid Contributor)
Brogues	Casual	13.6%	B-Tier (Mid Performer)
Loafers	Casual	9.7%	B-Tier (Mid Performer)
Sneakers	Casual	9.6%	C-Tier (Growth Opportunity)

Category-Level Performance

- Casual dominates in volume (11.3K units) but Formal leads in revenue per unit due to higher price points
- Semi-Casual is significantly under-indexed (2.5K units) — a potential growth category with targeted marketing
- The Formal category's average selling price drives disproportionately higher revenue relative to unit volume

18. Customer Analytics

Customer analysis focused on spend behavior, geographic distribution, and purchase frequency to identify high-value segments and retention opportunities.

Customer Segmentation Findings

- Top 10 customers by spend account for a disproportionate share of total revenue — classic Pareto distribution
- High-value customers predominantly reside in Andhra Pradesh, Rajasthan, Tamil Nadu, and Gujarat
- Average order value is approximately ₹14,478 per transaction
- Customers with multiple orders show cross-category purchasing (both Casual and Formal products)

Online vs. Offline Channel Performance

- Both channels contribute significantly to overall revenue
- Online orders demonstrate higher average order values, likely due to premium product accessibility
- Offline orders show higher transaction frequency in Tier-2 and Tier-3 cities
- Opportunity: Converting high-frequency offline buyers to the online channel can improve margin efficiency

19. Regional Performance Analysis

Top 5 States by Profit

State	Total Profit
Rajasthan	₹2,690,350
Tamil Nadu	₹2,685,720
Gujarat	₹2,432,550
Karnataka	₹2,346,630
Bihar	₹2,222,020

Regional Concentration Risk

- Top 10 states contribute a majority of total revenue — indicating over-reliance on specific markets
- Andaman & Nicobar Islands (15 orders, ₹42,300 profit) and Chandigarh (16 orders, ₹42,640) represent untapped micro-markets
- States like Assam (63 orders) and Chhattisgarh (142 orders) show nascent but growing demand
- Southern markets (Tamil Nadu, Karnataka, Andhra Pradesh) collectively dominate both revenue and profit

Strategic Opportunity Map

- High Priority Expansion: North-East India (Assam, Meghalaya) — low penetration but growing e-commerce adoption
- Growth Markets: Madhya Pradesh, Jharkhand, Odisha — mid-tier states with increasing disposable income
- Defend & Grow: Rajasthan, Tamil Nadu, Gujarat — top performers requiring loyalty programs and premium product launches

20. Sales Growth Analysis

- Overall annual sales growth of 4.76% reflects stable, consistent expansion
- The running revenue total shows a strong linear upward trend across the year
- Q1 (Jan-Mar) contributed the most to annual revenue due to the March spike
- Q3 (Jul-Sep) shows the fastest recovery trajectory — MoM growth from July trough to September peak
- Category-level growth: Formal category growing faster than Casual in revenue terms, driven by Single Monk premium pricing
- Geographic growth: States like Bihar (409 orders) and Delhi (236 orders) showing strong transaction velocity

Growth Insight: The 4.76% sales growth rate, while moderate, is achieved against a backdrop of seasonal demand volatility, regional concentration, and multi-channel complexity. This indicates strong underlying business fundamentals and significant runway for accelerated growth through targeted expansion, digital marketing investment, and product line diversification.

21. Profitability Analysis

Margin Performance by Category

- Overall profit margin of 22.4% is healthy for Indian retail footwear
- Formal category products (Single Monk, Oxford) command the highest margins due to premium pricing
- Casual category, despite high volume, operates on thinner margins
- Semi-Casual products show the most margin variability — opportunity for standardization

Cost Structure

- Total cost of ₹88M against ₹113M revenue implies an average cost-to-revenue ratio of 77.6%
- Cost per unit analysis reveals opportunities to renegotiate supplier terms for high-volume Casual products
- Online channel shows better cost efficiency due to reduced physical retail overhead

Profit Leakage Points

- Low-margin months (June, July) — fixed costs not offset by reduced sales volumes
- Underperforming states with high order counts but low profit margins suggest discount-heavy sales
- Semi-Casual category's low volume creates higher per-unit overhead

22. Recommendations

Revenue Growth

- Launch targeted digital marketing campaigns in March and August-September to amplify the natural seasonal peaks
- Introduce a loyalty rewards program for top 10% customers (accounting for the majority of revenue)
- Expand product availability in high-growth states: Bihar, Chhattisgarh, Assam
- Develop exclusive online-only bundles combining Casual and Formal products to increase basket size

Profitability Optimization

- Renegotiate procurement costs for high-volume Casual category products to improve margins by 2-3%
- Implement dynamic pricing for peak months (March, November) to capture additional margin
- Review and discontinue SKUs with consistently negative or near-zero profit margins
- Invest in reducing logistics costs for Tier-2/Tier-3 city deliveries

Market Expansion

- Prioritize North-East India for e-commerce channel expansion — low competition, growing demand
- Launch regional collections tailored to Southern states (AP, Tamil Nadu) which are the strongest markets
- Partner with regional distributors in underperforming states for offline channel expansion

Product Strategy

- Invest further in the Single Monk product line with new colors and variants — leverage its 35.3% revenue dominance
- Develop the Sneakers category with trend-forward designs to grow its 9.6% share
- Revitalize Semi-Casual category with marketing campaigns targeting young professionals

23. Future Improvements

- Predictive Sales Forecasting: Implement ARIMA or Prophet models in Python for 6-month revenue forecasting
- Real-Time Dashboard: Connect Power BI and Looker Studio to a live database (MySQL/PostgreSQL) for real-time KPI updates
- Customer RFM Segmentation: Build Recency-Frequency-Monetary segmentation model to classify customer tiers
- Cohort Analysis: Analyze customer retention and repeat purchase rates over time using cohort methodology
- Price Elasticity Modeling: Use regression analysis to determine optimal pricing for each product category
- AI Chatbot Integration: Embed a conversational AI layer into the dashboard for natural language data querying
- Supply Chain Analytics: Add inventory and procurement data to identify stockout risks and overstock patterns
- A/B Testing Framework: Design experiments to test pricing, promotional, and product placement strategies

24. Business Impact

This analytics platform delivers measurable business value across four dimensions:

Impact Area	Business Value Delivered
Decision Speed	Reduced report generation time from days to real-time self-service
Revenue Visibility	100% transparency into revenue drivers across 30 states
Margin Intelligence	Identified top and bottom margin products for pricing strategy
Market Targeting	Pinpointed 5+ high-opportunity expansion states
Customer Intelligence	Profiled top customers for loyalty and upsell programs
Operational Efficiency	Automated 15 recurring SQL reports replacing manual analysis
AI Efficiency	Generative AI reduced narrative report writing by 80%
Stakeholder Alignment	Executive-ready dashboards ensure data-driven boardroom discussions

25. AI Tools Used

AI Tool / Technology	Purpose	Integration Point
Generative AI (LLM API)	Auto-generate business narrative summaries from aggregated data	Python Jupyter Notebook
Claude (Anthropic)	Documentation assistance, SQL query optimization, insight generation	Project Planning & Reporting
ChatGPT (OpenAI)	DAX formula suggestions, dashboard layout ideation	Power BI Development Phase
Copilot (Microsoft)	Excel formula generation, pivot table automation	Excel Dashboard Phase
Python AI Libraries	Statistical analysis, outlier detection, visualization	EDA Notebook
AI-Assisted SQL Writing	Query structure validation and optimization suggestions	SQL Analytics Phase

26. Challenges Faced

- Metric Standardization: Aligning summary KPIs, rounded dashboard values, and detailed analytical tables across platforms required careful documentation control and reconciliation.
- DAX Complexity: Implementing time-intelligence measures (MoM growth, running totals) required careful understanding of filter context in Power BI
- Map Visualization: Ensuring Indian state names matched exactly to Bing Maps and Google Maps geographic boundaries for accurate choropleth rendering
- AI Prompt Engineering: Crafting effective prompts for Generative AI to produce accurate, domain-specific business narratives without hallucination
- Seasonal Bias: The extreme bimodal sales pattern (March peak vs. July trough) required careful interpretation to avoid misleading trend conclusions
- Multi-Platform Consistency: Maintaining visual and metric consistency across Power BI, Looker Studio, and Excel with different calculation engines

27. Learning Outcomes

Technical Skills Gained

- Advanced SQL: CTEs, Window Functions (LAG, RANK, SUM OVER), correlated subqueries
- Power BI: DAX time-intelligence, cross-filtering architecture, map visuals
- Python: Pandas feature engineering, statistical EDA, matplotlib/seaborn visualization
- Looker Studio: Google Maps integration, dynamic filtering, shareable report architecture
- AI Integration: Generative AI API calls, prompt engineering for business intelligence

Business Skills Gained

- Business storytelling through data visualization and narrative design
- KPI framework design for retail and e-commerce businesses
- Profitability analysis methodology and margin optimization thinking
- Geographic market analysis and regional expansion strategy formulation
- Executive communication: Translating complex data into boardroom-ready insights

28. Conclusion

The Wave Shoes Sales Analytics Dashboard is more than a data project — it is a complete, enterprise-grade business intelligence ecosystem that transforms raw transactional data into strategic intelligence, competitive advantage, and operational clarity.

This project demonstrates the power of integrating multiple analytics technologies — SQL for data wrangling, Python for statistical depth, Power BI and Looker Studio for visual storytelling, Excel for portable reporting, and Generative AI for intelligent narration — into a unified platform that serves stakeholders at every level of an organization.

The Wave Shoes analytics platform revealed that the business is on a healthy growth trajectory (4.76% growth, 22.4% margin), with clear seasonal patterns and product champions such as Single Monk. It also highlighted strategic opportunities in market expansion, category optimization, and digital-first customer acquisition.

As AI continues to reshape the analytics landscape, this project stands as a blueprint for how businesses can leverage data — not just to understand the past, but to engineer the future.

Project Repository github.com/deepak10281	Technologies Used Power BI Looker Studio MySQL
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Gen-AI-Powered-Sales-Analytics-Dashboard

Python | Excel | Generative AI